

Content-Aware Pseudo-Color Mapping for Scientific/Medical Grayscale Images

Advanced Digital Image and Video Processing (CSE 681)

Milestone 1

Project Repo: GitHub Repository

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Project Proposal & Literature Review

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Abstract

3–5 sentences: what problem are you tackling, why does it matter, what is your planned approach, and what public dataset/baseline will you use. Write this last.

[TODO: First Version abstract: revise after completing the implementation and evaluation experiments to reflect the actual methods and findings.]

1 Problem Statement & Motivation

Grayscale images are widely used in scientific and medical applications to represent information through variations in intensity. However, interpreting subtle intensity differences and identifying important image structures can be challenging, particularly when relevant features have intensity values similar to their surrounding regions. Pseudo-color mapping addresses this limitation by assigning colors to grayscale intensity levels, potentially improving visual differentiation and supporting image interpretation. Conventional approaches, such as intensity slicing using fixed Look-Up Tables (LUTs), assign colors according to predefined intensity-to-color relationships without considering the actual image content or the relative importance of different regions. Consequently, these approaches may not sufficiently emphasize task-relevant structures or regions of interest. This project investigates content-aware pseudo-color mapping, particularly saliency-guided mapping, as an alternative to conventional fixed-LUT techniques. The main objective is to investigate whether incorporating image saliency into the color-mapping process can improve the visual emphasis and localization of relevant features while maintaining desirable perceptual properties. The project is connected to Week 10 of the Digital Image Processing course, which covers pseudo-color processing, particularly HSI-based pseudo-color representation and intensity slicing. Building upon these concepts, the project extends conventional intensity-based color assignment by incorporating image-content information into the mapping process. Medical grayscale images, specifically chest X-ray images, will serve as the application domain for evaluating the proposed approach.

2 Related Work

Table 1: Key references.

Reference	Year	Relevance to this project
(Borland and Taylor, 2007)	2007	Explains perceptual pitfalls of rainbow colormaps.
(Zhou and Hansen, 2016)	2016	Surveys colormap generation and visualization design.
(Ware et al., 2023)	2023	Examines task-dependent strengths and limitations of rainbow colormaps.
(Selvaraju et al., 2020)	2020	Provides a foundation in visual explainability of deep learning models through gradient-based localization of image regions influencing predictions.
(Bujack et al., 2018)	2018	Provides a theoretical foundation for evaluating colormaps using perceptual uniformity, discriminability, and luminance properties.
(Qutbi, 2024)	2024	Demonstrates the application of perceptual metrics to quantitatively evaluate colormaps in medical-image visualization.
(Bylinskii et al., 2019)	2019	Reviews saliency evaluation metrics, supporting the selection of quantitative measures for ROI localization assessment.

Traditional pseudo-color mapping is a widely used image-processing technique that transforms grayscale images into color representations by assigning specific colors to different intensity levels. This process is commonly implemented using a Look-Up Table (LUT), in which each grayscale intensity is associated with a predefined RGB value. Consequently, for an 8-bit grayscale image, a LUT can contain 256 color entries, allowing each pixel to be mapped directly to its corresponding color without modifying the source intensity values. Different colormaps, such as Jet, Viridis, and other sequential color schemes, can be employed to represent intensity variations and improve the visual interpretation of image structures. However, the effectiveness of this technique strongly depends on the perceptual properties of the selected colormap. Borland and Taylor (2007) highlighted several limitations of the commonly used rainbow colormap, including non-uniform perceptual transitions and misleading visual gradients, which can negatively affect data interpretation. Zhou and Hansen (2016) further reviewed different colormap generation techniques and emphasized the relationship between colormap design and effective data visualization. More recently, Ware et al. (2023) revisited the limitations of rainbow colormaps and demonstrated that their effectiveness depends on the visualization task and the type of information being examined. Despite the advantages of fixed LUT-based mapping in providing a simple and computationally efficient visualization method, its color assignment depends exclusively on grayscale intensity rather than image content or regional importance. Therefore, pixels with identical intensities receive identical colors regardless of their anatomical location or relevance. This limitation motivates the investigation of content-aware pseudo-color mapping approaches that incorporate image saliency to adapt color representation and provide greater visual emphasis on potentially important regions.

The ChestX-ray8 paper Wang et al. (2017) provides the dataset foundation for our project, introducing a large-scale chest X-ray database with disease labels that will be used to train the

CNN-based classification model. The Grad-CAM paper Selvaraju et al. (2020) provides the explainability method, allowing us to generate saliency maps that highlight image regions contributing to the classifier's predictions. Together, these references support disease classification and saliency estimation, which form the basis of our proposed content-aware pseudo-color mapping approach.

Previous studies have established several perceptual properties that are important for evaluating the effectiveness of colormaps. Bujack et al. (2018) introduced a theoretical framework for assessing continuous colormaps, discussing properties such as perceptual uniformity, discriminability, and luminance behavior. These properties provide a foundation for evaluating whether a colormap represents grayscale intensity variations consistently and avoids misleading visual differences. In the context of medical imaging, Qutbi (2024) investigated the quantitative performance of commonly used colormaps using perceptual metrics, demonstrating the relevance of evaluating color representations according to their perceptual characteristics in medical-image visualization. Together, these studies motivate the use of intrinsic metrics to assess the perceptual quality of the proposed pseudo-color mapping against a conventional fixed-LUT baseline. Beyond intrinsic color properties, evaluating content-aware mapping requires investigating whether the emphasized regions correspond to task-relevant image structures. Bylinskii et al. (2019) examined different evaluation metrics for visual saliency models and discussed how their effectiveness depends on the characteristics of the saliency task and its ground-truth annotations. This provides a basis for selecting appropriate quantitative measures for evaluating saliency-based approaches. Building on these studies, the proposed work combines intrinsic colormap evaluation with ROI localization against available bounding-box annotations from the NIH Chest X-ray14 dataset. This allows the comparison to consider both the perceptual properties of the generated representations and their ability to emphasize annotated regions of interest.

3 Dataset & Baseline Plan

3.1 Dataset Description and Patient-Level Selection

We will use the publicly available NIH Chest X-ray dataset **?** distributed through Kaggle. The dataset contains 112,120 chest radiographs from 30,805 unique patients, provided as 1024×1024 images with image-level metadata, including patient identifiers, follow-up indices, view positions, and finding labels. The labels cover 14 thoracic conditions and a "No Finding" category; multiple findings may occur in one image.Kaggle lists the dataset under the CC0 1.0 public-domain dedication. We will retain the source documentation, acknowledge the dataset creators, make no attempt to re-identify patients, and present our results as visualization experiments rather than clinically validated diagnostic findings.

To reduce repeated observations of the same patient and keep the study manageable, we will retain only the first available image for each patient. Operationally, we will sort records within each patient by `Follow-up #` in ascending order, breaking ties by `Image Index`, and retain the first record. This is a reproducible index-based definition, not an independently verified acquisition date. Size is 30,805 images. The selected patient identifiers, filenames, and selection rule will be saved in a cohort manifest. A python script is made to automate this selection and save the manifest.

3.2 Training, Validation, and Test Protocol

The selected patients will be divided into 70% training, 20% validation, and 10% testing. We will shuffle a deterministically ordered patient list, assign $\lfloor 0.70N \rfloor$ patients to training and $\lfloor 0.20N \rfloor$

to validation, and assign the remainder to testing, where N is the verified cohort size. For $N = 30{,}805$, this gives 21,563 training, 6,161 validation, and 3,081 test images. Patient identifiers will be disjoint across the three splits, and the assignment will be saved rather than regenerated during each experiment.

Training data will be used for any learned components or fitted preprocessing parameters, validation data for selecting hyperparameters, and test data only for the final comparison. The fixed-LUT baseline itself requires no training. Both methods will be evaluated on the same held-out images; finding-label and view-position distributions will be reported for each split to make potential imbalances visible.

Table 2: Dataset summary.

Dataset	**Subset Used**	**Link**
NIH Chest X-rays	One image per patient: expected $N = 30{,}805$. Custom 70/20/10 patient-level split.	Kaggle dataset (CC0 1.0)

3.3 Fixed-LUT Viridis Baseline

We will compare the proposed saliency-aware mapping against a fixed Viridis lookup table. The baseline uses OpenCV's `cv2.applyColorMap` with `cv2.COLORMAP_VIRIDIS`: each 8-bit intensity in $[0, 255]$ is mapped to a fixed RGB color independently of image content. It introduces no learned parameters, saliency weighting or other content-aware adjustments. Native resolution and original pixel intensities are preserved.

An initial baseline module has already been implemented and run on 10 images from 10 different patients. Its outputs have been checked pixel-for-pixel against the fixed LUT. The baseline module accepts 8-bit grayscale PNGs, including grayscale stored as identical RGB channels with a fully opaque alpha channel. The module produces color PNGs, grayscale–Viridis comparison figures with an intensity colorbar, and a completed-run manifest recording input filenames, dimensions, library versions, and processing settings.

Figure 1 shows three initial baseline examples. These illustrate the fixed intensity-to-color mapping, not evidence of improved diagnostic performance.

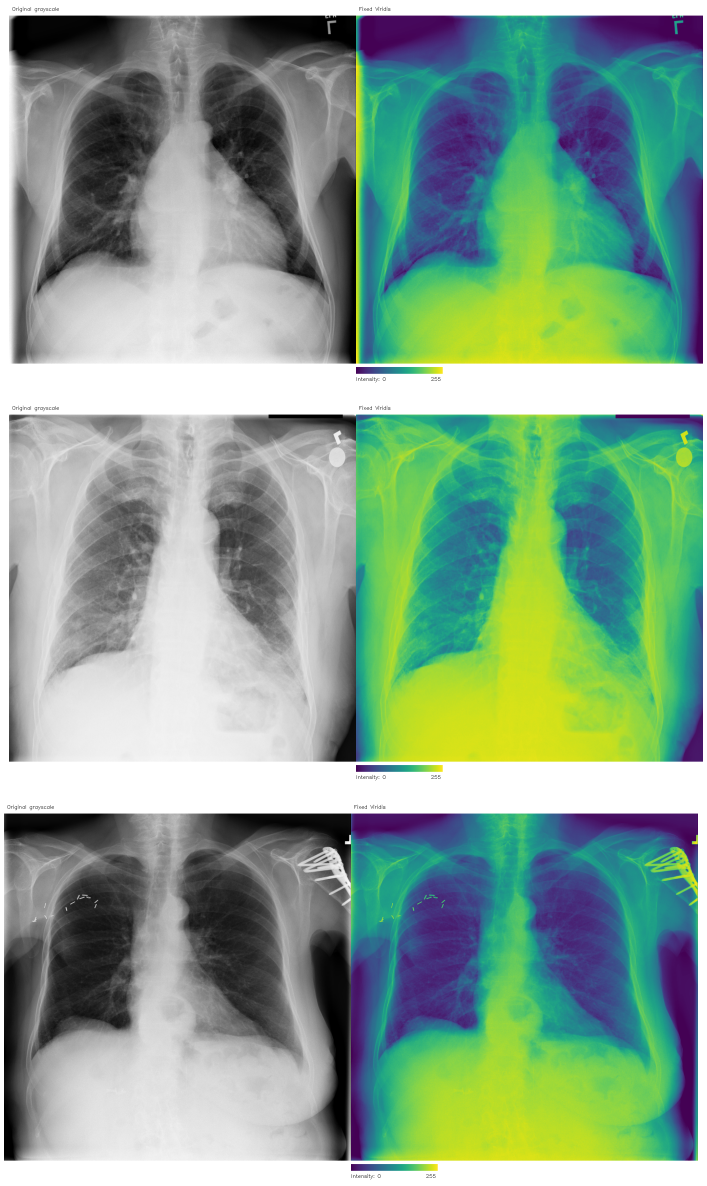


Figure 1: Initial fixed-LUT baseline samples for images `00000001_000.png`, `00000002_000.png`, and `00000003_000.png` (top to bottom). Each example shows the original grayscale image on the left and its fixed Viridis mapping on the right, with a colorbar for intensities from 0 to 255.

4 Proposed Method

4.1 Overview Architecture

Figure 2 illustrates the overall architecture of the proposed pseudo-color mapping framework, starting with the chest X-ray images, metadata, and patient-level cohort and split manifests stored in Google Drive. The images are imported, verified against the dataset records, cached locally, and preprocessed before being passed through two parallel processing paths. The first path implements the conventional fixed-LUT baseline, where the original grayscale intensities are directly mapped to colors using the Viridis colormap. The second path represents the proposed content-aware approach, in which a saliency estimation algorithm generates a saliency map that is subsequently combined with the original grayscale image to produce a content-aware colored representation. Finally, the outputs from both paths are forwarded to a common evaluation stage, where the original grayscale images, baseline results, and proposed outputs are visualized and compared using the defined evaluation criteria. This architecture enables a consistent comparison between conventional intensity-based mapping and the proposed saliency-guided approach using the same input images.

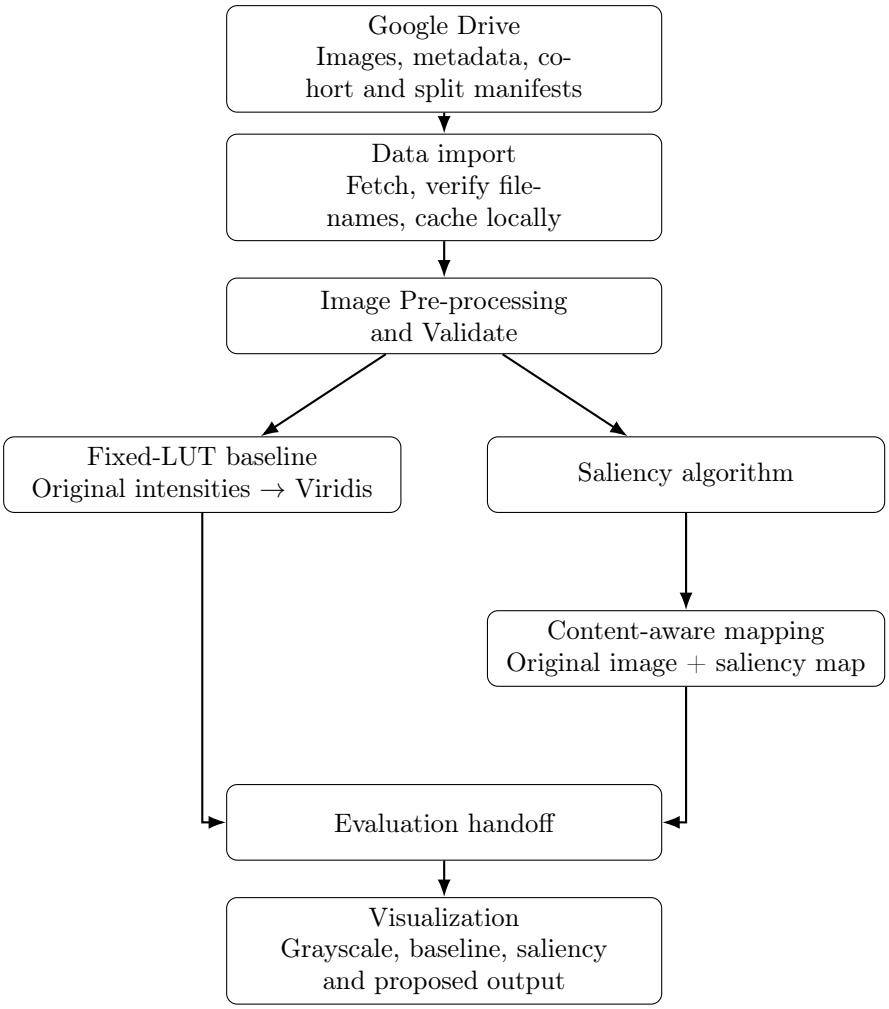


Figure 2: Overall architecture of the proposed content-aware pseudo-color mapping framework and its evaluation pipeline.

4.2 Saliency Estimation Process

The main objective of this part is to estimate the saliency of medical grayscale images, specifically chest X-ray images, by identifying the regions that contribute most to disease classification. To achieve this, we use a CNN-based classifier, specifically ResNet50, trained on the NIH ChestX-ray14 dataset to recognize different thoracic diseases. After training, Grad-CAM is applied to explain the model's predictions by highlighting the image regions that have the greatest influence on its decisions. The resulting saliency maps represent the spatial importance of different image regions and will be used as guidance for the next part of our project, which focuses on content-aware pseudo-color mapping.

Saliency Estimation Process:

4.2.1 CNN-Based Disease Classification

A CNN architecture, specifically ResNet50, is used to learn visual features from chest X-ray images.

The model is trained using disease labels to recognize patterns associated with different thoracic conditions.

The classification process consists of:

1. Extracting visual features through convolutional layers.
2. Learning disease-related patterns.
3. Producing prediction probabilities for the 14 disease categories.

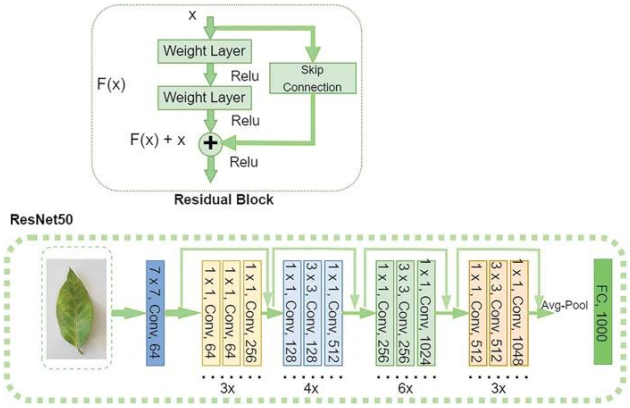


Figure 3: ResNet50 architecture used for chest X-ray disease classification.

4.2.2 Applying Grad-CAM for Explainability

After training the classifier, Grad-CAM is applied to understand which image regions contribute to its predictions.

Instead of simply producing a disease label, the technique generates a visual explanation of the model's decision.

Grad-CAM works by:

- Selecting a disease prediction to explain.

- Extracting feature maps from the final convolutional layer.
- Calculating gradients associated with the selected disease.
- Assigning importance weights to the feature maps.
- Combining the weighted feature maps to generate a heatmap.

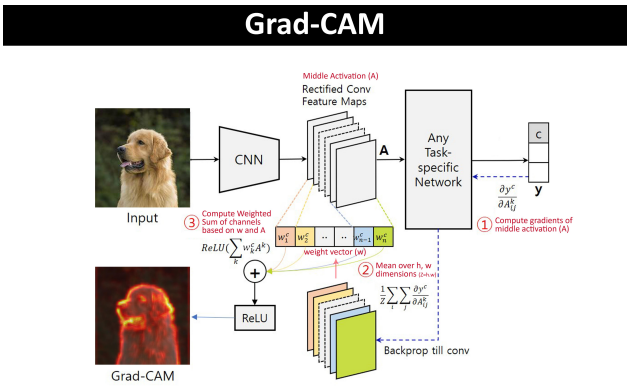


Figure 4: Grad-CAM workflow for generating visual explanations.

4.2.3 Saliency Map Generation

The Grad-CAM heatmap is resized to match the original X-ray dimensions and normalized to values between 0 and 1.

The resulting saliency map represents the spatial importance of different image regions.

- High saliency values indicate regions that strongly contribute to the selected classification.
- Low saliency values indicate regions with less positive contribution.

The generated map provides a spatial representation of the classifier's explanations and can guide subsequent image processing.

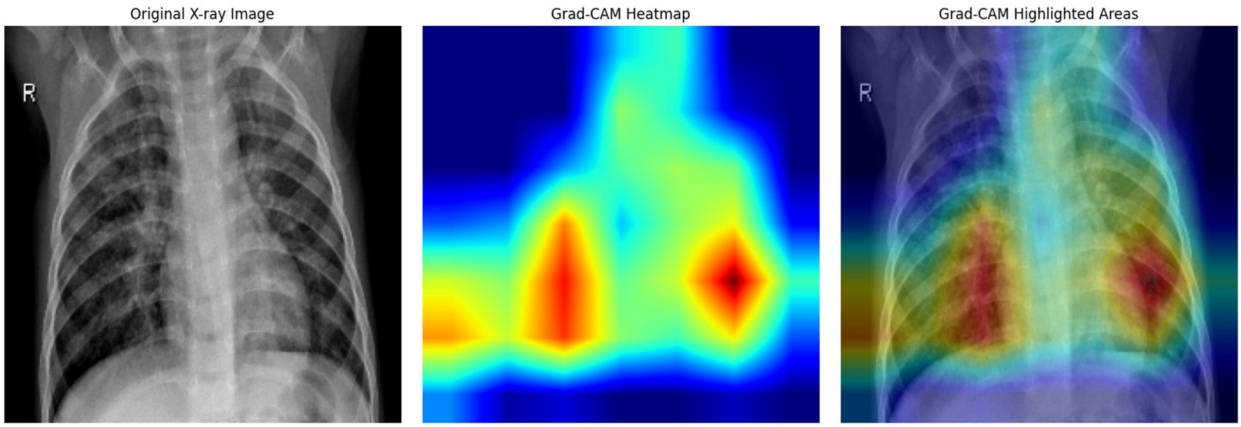


Figure 5: Original chest X-ray, Grad-CAM heatmap, and resulting saliency visualization.

4.3 Complete Workflow Diagram

The following diagram summarizes the complete workflow of the saliency estimation stage.

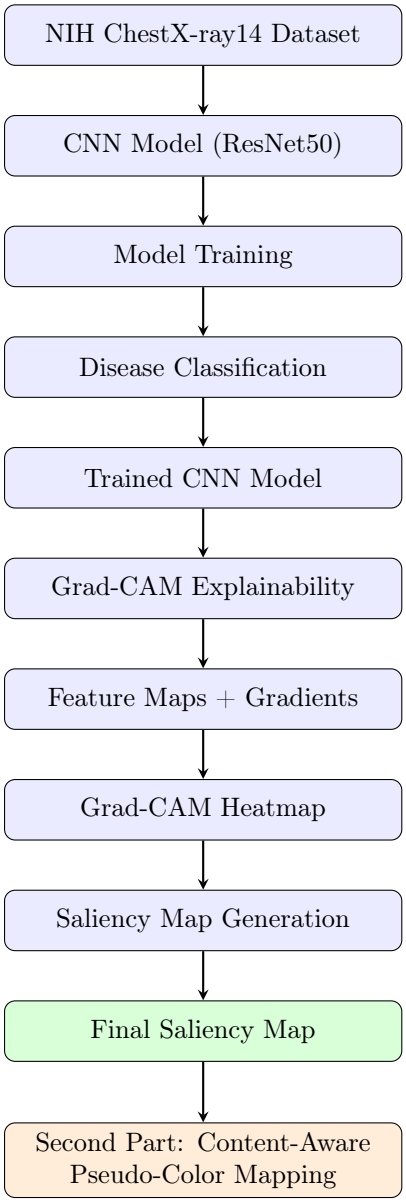


Figure 6: Complete workflow of the proposed saliency estimation stage.

At the end of this stage, we expect to obtain three main outputs:

Table 3: Expected outputs of the saliency estimation stage.

Output	**Description**
Trained CNN Classifier	A model capable of predicting thoracic disease categories.
Grad-CAM Heatmaps	Visual explanations highlighting regions contributing to disease predictions.
Saliency Maps	Normalized importance maps representing the spatial significance of image regions.

5 Feasibility & Compute Plan

The project is designed to be CPU-OK and feasible on a laptop-class machine without requiring a dedicated GPU. The pseudo-color mapping operations, including fixed-LUT mapping and saliency-guided color assignment, will primarily use CPU-based image processing with Python and standard image-processing libraries.

The initial fine-tuning experiments are estimated to require approximately 4 to 8 hours per experiment using CPU-based processing, assuming 5 to10 training epochs. Grad-CAM generation and subsequent pseudo-color mapping and evaluation operations will require additional processing time, depending on the number of images evaluated.

The proposed implementation and evaluation are therefore considered feasible within the project window using a laptop-class machine.

5.1 Minimum Viable Deliverable (MVP)

The minimum viable deliverable will consist of:

- Preprocessing a manageable subset of the NIH Chest X-ray14 dataset.
- Implementing conventional fixed-LUT pseudo-color mapping as the baseline.
- Implementing a saliency-guided content-aware pseudo-color mapping using Grad-CAM to estimate salient regions.
- Generating and comparing both representations using identical grayscale input images.
- Evaluating both mapping strategies using intrinsic color-mapping quality metrics and conducting ROI localization evaluation against the available ground-truth bounding-box annotations.

5.2 Stretch Goal

If the MVP is completed on schedule, the following additional objectives will be considered:

- Implementing and evaluating a grayscale-trained image-classification experiment using three input representations: grayscale, fixed-LUT pseudo-color, and content-aware pseudo-color images.

- Conducting a structured human-centered experiment to compare task accuracy, response time, perceived visibility, and participant confidence.
- Investigating different saliency-generation techniques or mapping configurations to determine their influence on the evaluation results.

6 Risk Assessment

The following risks have been identified, together with their proposed mitigation strategies:

- **Risk 1. Weakly Supervised Dataset Labels:** The NIH Chest X-ray14 dataset contains disease labels extracted from radiological reports using Natural Language Processing (NLP), with an expected accuracy of approximately 90%. Although these labels are reasonably accurate, some labeling errors may exist. Since disease classification is not the primary objective of this project, these labels will be treated as provided ground-truth labels without attempting to validate or correct them.
  Mitigation: Focus the evaluation on comparing the pseudo-color mapping approaches rather than disease classification accuracy. Clearly acknowledge the potential labeling inaccuracies as a dataset limitation.
- **Risk 2. Limited ROI Annotations:** Ground-truth bounding-box annotations are available for only a small subset of approximately 1,000 images in the NIH Chest X-ray14 dataset. This may limit the number of samples available for quantitative ROI localization evaluation.
  Mitigation/Fallback: Use the available annotated subset and clearly report its size and limitations. The ROI localization experiment will be treated as an evaluation on the annotated subset rather than the entire dataset.
- **Risk 3. Human-Study Participation:** Limited participant availability or inconsistent responses may affect the reliability of the human-centered evaluation.
  Mitigation/Fallback: Prepare a simple structured experiment with predefined visual tasks and consistent presentation conditions. If participant recruitment becomes impractical, prioritize the quantitative evaluation methods included in the MVP.

7 Work Breakdown

The project is divided among three students according to the main development and evaluation responsibilities. The tasks are organized into three milestones to ensure that dataset preparation, algorithm development, and evaluation progress in parallel.

All team members will contribute to integration, comparative analysis, interpretation of results, and final report preparation

Table 4: Milestone task allocation.

Milestone	Student A	Student B	Student C
M1 (Wk 3)	Dataset investigation, data requirements, and baseline research.	Literature review and investigation of content-aware mapping approaches.	Literature review and development of the evaluation framework and criteria.
M2 (Wk 6)	Dataset downloading, preprocessing, splitting, and implementation of the fixed-LUT baseline.	Implementation of the proposed saliency-guided content-aware mapping approach.	Implementation of intrin sic evaluation metrics and ROI localization evaluation.
M3 (Wk 9)	Dataset preparation sup port, baseline validation, and classification data preparation.	Finalization and validation of the proposed mapping algorithm and generated outputs.	Downstream-task evaluation, analysis of quantitative results, and human centered evaluation if feasible.

References

David Borland and Russell M. Taylor, II. Rainbow color map (still) considered harmful. *IEEE Computer Graphics and Applications*, 27(2):14–17, 2007. doi: 10.1109/MCG.2007.323435.

Liang Zhou and Charles D. Hansen. A survey of colormaps in visualization. *IEEE Transactions on Visualization and Computer Graphics*, 22(8):2051–2069, 2016. doi: 10.1109/TVCG.2015.2489649.

Colin Ware, Maureen Stone, and Danielle Albers Szafir. Rainbow colormaps are not all bad. *IEEE Computer Graphics and Applications*, 43(3):88–93, 2023. doi: 10.1109/MCG.2023.3246111.

Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-CAM: Visual explanations from deep networks via gradient-based localization. *International Journal of Computer Vision*, 128:336–359, 2020. doi: 10.1007/s11263-019-01228-7.

Robert Bujack, Thomas L. Turton, Francesca Samsel, Colin Ware, David H. Rogers, and James Ahrens. The good, the bad, and the ugly: A theoretical framework for the assessment of continuous colormaps. *IEEE Transactions on Visualization and Computer Graphics*, 24(1):923–933, 2018. doi: 10.1109/TVCG.2017.2743978.

Mehrdad Qutbi. Quantitative performance evaluation of commonly used colormaps for image display in myocardial perfusion imaging: Analysis based on perceptual metrics. *Molecular Imaging and Radionuclide Therapy*, 33(2):94–105, 2024. doi: 10.4274/mirt.galenos.2024.34711.

Zoya Bylinskii, Tilke Judd, Aude Oliva, Antonio Torralba, and Frédo Durand. What do different evaluation metrics tell us about saliency models? *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(3):740–757, 2019. doi: 10.1109/TPAMI.2018.2815601.

Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M. Summers. Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 2097–2106, 2017. doi: 10.1109/CVPR.2017.369.